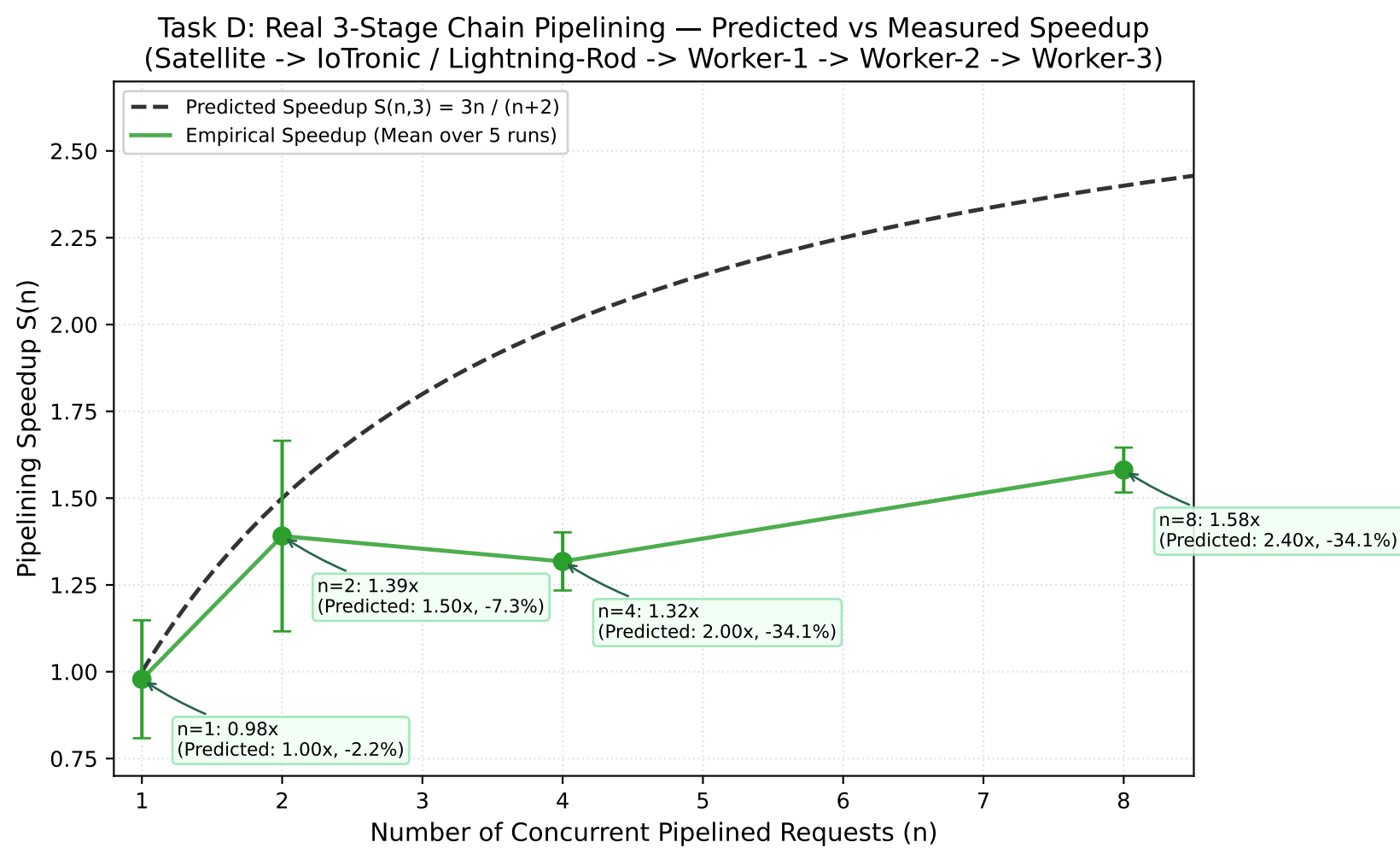
Task D: Real 3-Stage Chain Pipelining — Predicted vs Measured Speedup
(Satellite -> IoTronic / Lightning-Rod -> Worker-1 -> Worker-2 -> Worker-3)
Predicted Speedup S(n,3) = 3n / (n+2)
Empirical Speedup (Mean over 5 runs)
Pipelining Speedup S(n)
Number of Concurrent Pipelined Requests (n)
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0.75
1
2
3
4
5
6
7
8
n=1: 0.98x
(Predicted: 1.00x, -2.2%)
n=2: 1.39x
(Predicted: 1.50x, -7.3%)
n=4: 1.32x
(Predicted: 2.00x, -34.1%)
n=8: 1.58x
(Predicted: 2.40x, -34.1%)